Assessment Menu: Teacher Guide

The Rising Soufflé Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
GRAPH	Graphing parabolas, transformations and the forms of a quadratic	A1, A2, E1, S1
SOLVE	Solving quadratic equations and the discriminant	A3, A4, E2, E3
COMPLEX	Imaginary numbers, complex arithmetic and complex roots	A5, E4, E5, S2
APPLY	Quadratic models, inequalities and systems	A6, E6, S3, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

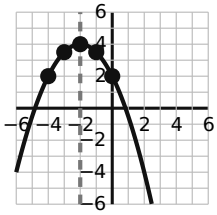
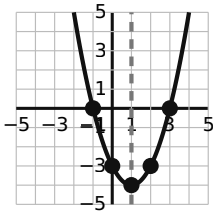
Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

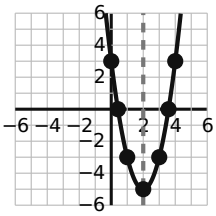
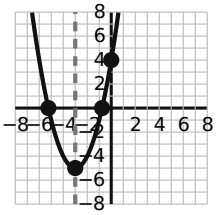
Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Answer graphs are drawn from the functions themselves.

Dish	Problem	Solution	Rubric
A1 GRAPH	Let $f(x) = -\frac{1}{2}(x+2)^2 + 4$. **(a)** Give the vertex, the axis of symmetry and the direction of opening. **(b)** Describe the transformations of $y = x^2$. **(c)** Find the y-intercept. **(d)** Graph $f(x)$.	**Answer:** (a) vertex $(-2, 4)$, axis $x = -2$, opens down (maximum value 4) (b) reflect over the x-axis, vertical compression by $\frac{1}{2}$, left 2, up 4 (c) (0, 2) (d) graph on the grid (a) Vertex form $a(x-h)^2 + k$ with $h = -2$ and $k = 4$: **vertex (−2, 4)**, **axis x = −2**. Since $a = -\frac{1}{2} < 0$, it **opens down**. (b) Reflect $y = x^2$ over the x-axis, compress vertically by $\frac{1}{2}$, shift **left 2** ($x + 2$ means $h = -2$) and up 4. (c) $f(0) = -\frac{1}{2}(4) + 4 = \mathbf{2}$, so (0, 2). (d) Points: $(-2, 4)$, $(-3, \frac{7}{2})$, $(-1, \frac{7}{2})$, (0, 2), (−4, 2). The parabola passes through them; (0, 2) and (−4, 2) mirror across $x = -2$. 	**[1]** (a) Vertex $(-2, 4)$ and axis $x = -2$ (sign of h correct) **[1]** (a) Opens down because a is negative **[1]** (b) Reflection, compression by $\frac{1}{2}$, left 2 and up 4 **[1]** (c) y-intercept (0, 2) **[1]** (d) Parabola graphed through the vertex and symmetric points *[n] = points*
A2 GRAPH	Graph $y = (x+1)(x-3)$. **(a)** Find the x-intercepts. **(b)** Find the axis of symmetry and the vertex. **(c)** Write the equation in standard form and find the y-intercept.	**Answer:** (a) $(-1, 0)$ and $(3, 0)$ (b) axis $x = 1$, vertex $(1, -4)$ (c) $y = x^2 - 2x - 3$, y-intercept $(0, -3)$ (graph on the grid) (a) Set each factor to 0: $x = -1$ and $x = 3$, so **(−1, 0) and (3, 0)**. (b) The axis is halfway between the intercepts: $x = \frac{-1+3}{2} = \mathbf{1}$. Then $y = (2)(-2) = -4$: **vertex (1, −4)**. (c) $(x+1)(x-3) = x^2 - 3x + x - 3 = \mathbf{x^2 - 2x - 3}$; at $x = 0$, $y = \mathbf{-3}$. Graph: plot the intercepts, vertex and y-intercept; (2, −3) mirrors (0, −3). The parabola opens up. 	**[1]** (a) Both x-intercepts **[1]** (b) Axis $x = 1$ (midpoint of the intercepts) **[1]** (b) Vertex $(1, -4)$ **[1]** (c) Standard form $x^2 - 2x - 3$ and y-intercept $(0, -3)$ **[1]** Parabola graphed through all of these points *[n] = points*
A3 SOLVE	Solve each equation. **(a)** $3x^2 - 48 = 0$ **(b)** $(x-5)^2 = 36$ **(c)** $x^2 + 2x - 15 = 0$ **(d)** $2x^2 = 5x$. In (d), why is dividing both sides by x a mistake?	**Answer:** (a) $x = \pm 4$ (b) $x = 11$ or $x = -1$ (c) $x = 3$ or $x = -5$ (d) $x = 0$ or $x = \frac{5}{2}$ (dividing by x loses the solution $x = 0$) (a) $3x^2 = 48$, $x^2 = 16$, so **$x = \pm 4$**. (b) $x - 5 = \pm 6$, so **$x = 11$** or **$x = -1$**. (c) $(x+5)(x-3) = 0$: **$x = -5$** or **$x = 3$**. (d) $2x^2 - 5x = 0$, $x(2x-5) = 0$: **$x = 0$** or **$x = \frac{5}{2}$**. Dividing by x throws away the solution $x = 0$.	**[1]** (a) Both solutions ± 4 **[1]** (b) Takes ± of the square root and solves both **[1]** (c) Factors correctly; $x = 3$ and $x = -5$ **[2]** (d) $x = 0$ and $x = \frac{5}{2}$ (factors out x instead of dividing) *[n] = points*
A4 SOLVE	Find $b^2 - 4ac$ and describe the solutions (how many, real or nonreal, rational or irrational). **(a)** $2x^2 - 3x - 5 = 0$ **(b)** $4x^2 - 12x + 9 = 0$ **(c)** $x^2 - 2x + 5 = 0$. **(d)** Find k so that $x^2 + 6x + k = 0$ has exactly one real solution.	**Answer:** (a) 49: two rational real solutions (b) 0: one repeated rational solution (c) −16: two nonreal (complex) solutions (d) $k = 9$ (a) $(-3)^2 - 4(2)(-5) = 9 + 40 = \mathbf{49}$ (a perfect square): two rational real solutions. (b) $(-12)^2 - 4(4)(9) = 144 - 144 = \mathbf{0}$: one repeated real solution. (c) $(-2)^2 - 4(1)(5) = 4 - 20 = \mathbf{-16}$: no real solutions, two complex conjugates. (d) $36 - 4k = 0$, so **$k = 9$**.	**[1]** (a) 49 (square of the negative b done correctly) and two rational solutions **[1]** (b) 0 and one repeated solution **[1]** (c) −16 and two nonreal solutions **[2]** (d) Sets $36 - 4k = 0$ and $k = 9$ *[n] = points*
A5 COMPLEX	Simplify. Write complex numbers in $a + bi$ form. **(a)** $\sqrt{-72}$ **(b)** i^{23} **(c)** $(3 - 2i) + (-5 + 6i)$ **(d)** $(4 + i)(4 - i)$	**Answer:** (a) $6i\sqrt{2}$ (b) $-i$ (c) $-2 + 4i$ (d) 17 (a) $\sqrt{-72} = i\sqrt{72} = i \cdot \sqrt{36 \cdot 2} = \mathbf{6i\sqrt{2}}$. (b) $i^{23} = (i^4)^5 \cdot i^3 = 1 \cdot (-i) = \mathbf{-i}$. (c) Combine real and imaginary parts: **$-2 + 4i$**. (d) $16 - i^2 = 16 - (-1) = \mathbf{17}$.	**[1]** (a) $6i\sqrt{2}$ **[1]** (b) $-i$ (uses the cycle of 4) **[1]** (c) $-2 + 4i$ **[2]** (d) Uses $i^2 = -1$ to get 17 (not 15) *[n] = points*
A6 APPLY	Solve $x^2 - x - 6 > 0$. **(a)** Graph the solution on the number line. **(b)** Write it in interval notation. **(c)** Write the solution of $x^2 - x - 6 \le 0$ in interval notation.	**Answer:** (a) open circles at −2 and 3, shade left of −2 and right of 3 (b) $(-\infty, -2) \cup (3, \infty)$ (c) $[-2, 3]$ Factor: $(x-3)(x+2) = 0$, boundary points **$x = -2$** and **$x = 3$**. Test $x = 0$: $-6 > 0$ is false, so the middle is **not** in the solution; test $x = 5$: $14 > 0$ is true. (a)–(b) Open circles (strict inequality); **$(-\infty, -2) \cup (3, \infty)$**. (c) The opposite region, with closed endpoints: **$[-2, 3]$**.	**[1]** Boundary points −2 and 3 by factoring **[1]** Tests a point in each region **[1]** (a) Open circles with the correct two regions shaded **[1]** (b) $(-\infty, -2) \cup (3, \infty)$ **[1]** (c) $[-2, 3]$ *[n] = points*

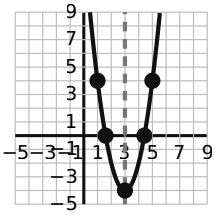
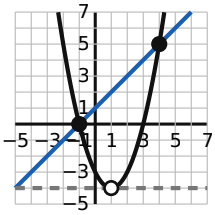
Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Answer graphs are drawn from the functions themselves.

Dish	Problem	Solution	Rubric
E1 GRAPH	Let $f(x) = 2x^2 - 8x + 3$. **(a)** Complete the square to write $f(x)$ in vertex form. **(b)** Name the vertex and the axis of symmetry. **(c)** Find the x-intercepts exactly. **(d)** Graph $f(x)$ and label the vertex and intercepts.	**Answer:** (a) $f(x) = 2(x-2)^2 - 5$ (b) vertex $(2, -5)$, axis $x = 2$ (c) $x = \frac{4 \pm \sqrt{10}}{2}$ (d) graph on the grid (y-intercept (0, 3)) (a) $2(x^2 - 4x) + 3 = 2(x^2 - 4x + 4) - 8 + 3$, so $\mathbf{f(x) = 2(x-2)^2 - 5}$. (b) **Vertex (2, −5)**, **axis x = 2**; it opens up. (c) $2(x-2)^2 = 5$, $(x-2)^2 = \frac{5}{2}$, $x - 2 = \pm\frac{\sqrt{10}}{2}$: $\mathbf{x = \frac{4 \pm \sqrt{10}}{2}}$. (d) Plot (2, −5), (0, 3), (4, 3), (1, −3), (3, −3) and the intercepts; draw the parabola. 	**[2]** (a) Factors out 2 from the x terms **[2]** (a) Completes the square and subtracts 8 to balance **[1]** (a) $f(x) = 2(x-2)^2 - 5$ **[1]** (b) Vertex (2, −5) and axis $x = 2$ **[2]** (c) Solves $2(x-2)^2 - 5 = 0$ correctly, with ± **[2]** (d) Vertex, intercepts and symmetric points on a smooth parabola *[n] = points*
E2 SOLVE	**(a)** Solve $x^2 + 6x + 4 = 0$ by completing the square. **(b)** Solve $2x^2 - 12x + 7 = 0$ by completing the square (watch the leading coefficient). **(c)** Graph $y = x^2 + 6x + 4$ and plot the x-intercepts from (a).	**Answer:** (a) $x = -3 \pm \sqrt{5}$ (b) $x = \frac{6 \pm \sqrt{22}}{2}$ (c) vertex $(-3, -5)$; x-intercepts about −5.24 and −0.76 (a) $x^2 + 6x = -4$; add 9 to both sides: $(x+3)^2 = 5$, so $\mathbf{x = -3 \pm \sqrt{5}}$. (b) Divide by 2: $x^2 - 6x = -\frac{7}{2}$. Add 9 to both sides: $(x-3)^2 = \frac{11}{2}$, $x - 3 = \pm\frac{\sqrt{22}}{2}$, so $\mathbf{x = \frac{6 \pm \sqrt{22}}{2}}$. (c) $y = (x+3)^2 - 5$: vertex $(-3, -5)$, y-intercept $(0, 4)$; it crosses the x-axis near −5.24 and −0.76, matching (a). 	**[1]** (a) Moves the constant and adds 9 to both sides **[1]** (a) $(x+3)^2 = 5$ **[1]** (a) $x = -3 \pm \sqrt{5}$ **[1]** (b) Divides every term by 2 first **[1]** (b) Adds 9 to both sides **[1]** (b) $(x-3)^2 = \frac{11}{2}$ **[1]** (b) $x = (6 \pm \sqrt{22})$ over 2 **[1]** (c) Vertex $(-3, -5)$ plotted **[1]** (c) Smooth parabola with y-intercept (0, 4) **[1]** (c) x-intercepts plotted and compared with (a) *[n] = points*
E3 SOLVE	**(a)** Use the quadratic formula on $3x^2 - 4x - 2 = 0$. Give exact solutions, then decimals to the nearest hundredth. **(b)** Solve $2x^2 - 6x + 5 = 0$. **(c)** A student wrote $\frac{4 \pm 2\sqrt{10}}{6} = 2 \pm \frac{\sqrt{10}}{3}$. Find the error.	**Answer:** (a) $x = \frac{2 \pm \sqrt{10}}{3}$, so $x = 1.72$ or $x = -0.39$ (b) $x = \frac{3 \pm i}{2}$ (c) the 2 in the numerator is divided only in the radical term; every term must be divided (a) $a = 3, b = -4, c = -2$: $x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(3)(-2)}}{2(3)} = \frac{4 \pm 2\sqrt{10}}{6} = \mathbf{\frac{2 \pm \sqrt{10}}{3}}$. Decimals: **1.72** and **−0.39**. (b) Discriminant: $36 - 40 = -4$. $x = \frac{6 \pm \sqrt{-4}}{4} = \mathbf{\frac{3 \pm i}{2}}$. (c) Every term of the numerator must be divided by 2. The student divided the 4 but left the $2\sqrt{10}$ term undivided.	**[1]** (a) Substitutes $a = 3$, $b = -4$, $c = -2$ with correct signs **[1]** (a) Discriminant 40 **[2]** (a) Simplified exact form $(2 \pm \sqrt{10})$ over 3 **[1]** (a) 1.72 and −0.39 **[1]** (b) Discriminant −4 **[2]** (b) $(3 \pm i)$ over 2 **[2]** (c) Explains that every term of the numerator must be divided *[n] = points*
E4 COMPLEX	Write each answer in $a + bi$ form. **(a)** $(5 - 3i) - (-2 + 4i)$ **(b)** $(2 + 3i)(4 - 5i)$ **(c)** $(3 - 2i)^2$ **(d)** Find $\lvert 6 - 8i \rvert$, the distance from the origin in the complex plane.	**Answer:** (a) $7 - 7i$ (b) $23 + 2i$ (c) $5 - 12i$ (d) 10 (a) $5 - 3i + 2 - 4i = \mathbf{7 - 7i}$. (b) $8 - 10i + 12i - 15i^2 = 8 + 2i + 15 = \mathbf{23 + 2i}$. (c) $9 - 12i + 4i^2 = 9 - 4 - 12i = \mathbf{5 - 12i}$. (d) $\sqrt{6^2 + (-8)^2} = \sqrt{100} = \mathbf{10}$.	**[2]** (a) Distributes the subtraction $(7 - 7i$, not $3 + i)$ **[3]** (b) FOIL, uses $i^2 = -1$, and $23 + 2i$ **[3]** (c) Expands all three terms (not just $9 + 4i^2$) and gets $5 - 12i$ **[2]** (d) $\sqrt{(36 + 64)} = 10$ *[n] = points*
E5 COMPLEX	**(a)** Write $\frac{10 + 5i}{2 - i}$ in $a + bi$ form. **(b)** Write $\frac{3 + 2i}{1 + 4i}$ in $a + bi$ form. **(c)** Find $(3 - 5i)(3 + 5i)$ and explain why multiplying by a conjugate helps with division.	**Answer:** (a) $3 + 4i$ (b) $\frac{11}{17} - \frac{10}{17}i$ (c) 34; the product of conjugates is a real number, so the denominator becomes real (a) Multiply top and bottom by $2 + i$: numerator $20 + 10i + 10i + 5i^2 = 15 + 20i$; denominator $4 - i^2 = 5$. So $\frac{15 + 20i}{5} = \mathbf{3 + 4i}$. (b) Multiply by $1 - 4i$: numerator $3 - 12i + 2i - 8i^2 = 11 - 10i$; denominator $1 - 16i^2 = 17$. So $\mathbf{\frac{11}{17} - \frac{10}{17}i}$. (c) $9 - 25i^2 = 9 + 25 = \mathbf{34}$. A complex number times its conjugate is real, so the denominator of a quotient becomes real.	**[1]** (a) Multiplies by the conjugate $2 + i$ (top and bottom) **[1]** (a) Numerator $15 + 20i$, denominator 5 **[2]** (a) $3 + 4i$ **[1]** (b) Multiplies by the conjugate $1 - 4i$ **[1]** (b) Numerator $11 - 10i$ **[1]** (b) Denominator 17 **[1]** (c) 34 **[2]** (c) Explains that the denominator becomes real *[n] = points*
E6 APPLY	A chef tosses pizza dough from 6 ft. Its height is $h(t) = -16t^2 + 40t + 6$ feet after t seconds. **(a)** Find the maximum height and when it occurs. **(b)** When does it hit the floor, to the nearest hundredth of a second? **(c)** When is it 20 ft high? Explain the two answers.	**Answer:** (a) maximum 31 ft at $t = \frac{5}{4}$ s (1.25 s) (b) $t = \frac{5 + \sqrt{31}}{4} = 2.64$ s (the negative root is rejected) (c) $t = \frac{5 - \sqrt{11}}{4} = 0.42$ s on the way up and $t = \frac{5 + \sqrt{11}}{4} = 2.08$ s on the way down (a) $t = \frac{-40}{2(-16)} = \frac{5}{4}$ s. $h(1.25) = -16 \cdot \frac{25}{16} + 50 + 6 = -25 + 56 = \mathbf{31\ ft}$. The max height is the y-value, 31, not 1.25. (b) $-16t^2 + 40t + 6 = 0$, or $8t^2 - 20t - 3 = 0$: $t = \frac{5 \pm \sqrt{31}}{4}$. The other root is about −0.14 s (before the toss): reject it, so $\mathbf{t = 2.64}$ **s**. (c) $-16t^2 + 40t - 14 = 0$, or $8t^2 - 20t + 7 = 0$: $t = \frac{5 \pm \sqrt{11}}{4}$, so **0.42 s and 2.08 s**. Once going up, once coming down.	**[1]** (a) $t = 1.25$ s **[1]** (a) Maximum height 31 ft (the y-value) **[1]** (a) Units on both answers **[1]** (b) Sets $h(t) = 0$ **[1]** (b) Correct quadratic formula work **[1]** (b) 2.64 s with the negative root rejected **[1]** (c) Sets $h(t) = 20$ and gets $8t^2 - 20t + 7 = 0$ **[1]** (c) Correct formula work **[1]** (c) Both times, 0.42 s and 2.08 s **[1]** (c) Explains going up and coming down *[n] = points*

Answer Key: Chef's Specials (20 pts each)

Worked solutions and rubrics. Answer graphs are drawn from the functions themselves.

Dish	Problem	Solution	Rubric
S1 GRAPH	A parabola has vertex (3, −4) and passes through (1, 4). **(a)** Write its equation in vertex form. **(b)** Write it in standard form. **(c)** Find the x-intercepts exactly and the y-intercept. **(d)** Graph it, plotting the vertex, the given point, its mirror image and the x-intercepts. **(e)** Give the axis of symmetry and the mirror image of (1, 4).	**Answer:** (a) $y = 2(x-3)^2 - 4$ (b) $y = 2x^2 - 12x + 14$ (c) $x = 3 \pm \sqrt{2}$; y-intercept (0, 14) (e) axis $x = 3$; mirror image (5, 4) (a) Substitute (1, 4) into $y = a(x-3)^2 - 4$: $4 = a(4) - 4$, so $8 = 4a$ and $a = 2$. **$y = 2(x-3)^2 - 4$**. (b) $2(x^2 - 6x + 9) - 4 = 2x^2 - 12x + 18 - 4 = \mathbf{2x^2 - 12x + 14}$. (c) $2(x-3)^2 = 4$, $(x-3)^2 = 2$, so **$x = 3 \pm \sqrt{2}$** (about 1.59 and 4.41). At $x = 0$, $y =$ **14**, off this grid. (d) Plot (3, −4), (1, 4), (5, 4) and the x-intercepts; connect with a smooth, upward-opening curve. (e) **Axis $x = 3$**; the mirror image of (1, 4) is **(5, 4)**. 	**[2]** (a) Substitutes (1, 4) into the vertex form **[2]** (a) Solves $a = 2$ **[1]** (a) $y = 2(x-3)^2 - 4$ **[2]** (b) Expands $(x-3)^2$ correctly **[1]** (b) Distributes the 2 and combines constants **[1]** (b) $y = 2x^2 - 12x + 14$ **[1]** (c) Sets $y = 0$ **[1]** (c) $(x-3)^2 = 2$ **[2]** (c) $x = 3 \pm \sqrt{2}$ **[1]** (c) y-intercept (0, 14) **[1]** (d) Vertex and (1, 4) plotted **[1]** (d) (5, 4) plotted **[1]** (d) x-intercepts plotted **[1]** (d) Smooth parabola **[1]** (e) Axis $x = 3$ **[1]** (e) (5, 4) *[n] = points*
S2 COMPLEX	**(a)** Solve $x^2 - 6x + 25 = 0$ with the quadratic formula. **(b)** Show the two roots are conjugates, and find their sum and product. **(c)** Write a quadratic with integer coefficients whose roots are $2 + 3i$ and $2 - 3i$. **(d)** Verify by substitution that $3 + 4i$ is a root of $x^2 - 6x + 25 = 0$. **(e)** For what values of k does $x^2 + 2x + k = 0$ have two nonreal solutions?	**Answer:** (a) $x = 3 \pm 4i$ (b) conjugates; sum 6, product 25 (c) $x^2 - 4x + 13 = 0$ (d) the expression equals 0 (e) $k > 1$ (a) Discriminant $36 - 100 = -64$. $x = \dfrac{6 \pm \sqrt{-64}}{2} = \dfrac{6 \pm 8i}{2} = \mathbf{3 \pm 4i}$. (b) $3 + 4i$ and $3 - 4i$ differ only in the sign of the imaginary part. Sum = **6** ($= -b$); product $= 9 - 16i^2 =$ **25** ($= c$). (c) $[x - (2+3i)][x - (2-3i)] = (x-2)^2 - (3i)^2 = x^2 - 4x + 4 + 9 = \mathbf{x^2 - 4x + 13 = 0}$. (d) $(3+4i)^2 = 9 + 24i - 16 = -7 + 24i$; $-6(3+4i) = -18 - 24i$. Total: $-7 + 24i - 18 - 24i + 25 =$ **0** ✓. (e) Discriminant $4 - 4k < 0$, so **$k > 1$**.	**[1]** (a) Discriminant −64 **[1]** (a) $\sqrt{-64} = 8i$ **[1]** (a) Divides the whole numerator by 2 **[2]** (a) $3 \pm 4i$ **[1]** (b) Identifies the conjugate pair **[1]** (b) Sum 6 **[1]** (b) Product 25, using $i^2 = -1$ **[2]** (c) Builds $[x - (2+3i)][x - (2-3i)]$ **[2]** (c) $x^2 - 4x + 13 = 0$ (+9, not −9) **[2]** (d) Squares $3 + 4i$ correctly: $-7 + 24i$ **[1]** (d) $-6(3+4i) = -18 - 24i$ **[2]** (d) Combines to 0 **[1]** (e) Discriminant $4 - 4k$ **[1]** (e) Sets it less than 0 **[1]** (e) $k > 1$ *[n] = points*
S3 APPLY	The bistro sells 120 bowls of soup a day at $8. Each $1 price increase loses 10 bowls. Let x be the number of $1 increases. **(a)** Write the daily revenue $R(x)$ in standard form. **(b)** Find the price that maximizes revenue and the maximum revenue. **(c)** For what prices is revenue at least $960? **(d)** State a sensible domain and the range of $R(x)$.	**Answer:** (a) $R(x) = (8 + x)(120 - 10x) = -10x^2 + 40x + 960$ (b) price $10, maximum revenue $1000 (at $x = 2$, 100 bowls) (c) prices from $8 to $12 (d) domain $0 \le x \le 12$; range $0 \le R(x) \le 1000$ (a) Price $8 + x$, bowls $120 - 10x$: $R(x) = (8 + x)(120 - 10x) = 960 - 80x + 120x - 10x^2 = \mathbf{-10x^2 + 40x + 960}$. (b) $x = \dfrac{-40}{2(-10)} = 2$ increases: price $8 + 2 =$ **$10**, bowls $120 - 20 = 100$. $R(2) = -40 + 80 + 960 =$ **$1000**. The maximum revenue is the output, not $x = 2$. (c) $-10x^2 + 40x + 960 \ge 960$ gives $-10x(x-4) \ge 0$, so $0 \le x \le 4$ (test $x = 2$: $80 \ge 0$). Prices $8 + x$ range **from $8 to $12**. (d) $x \ge 0$ (increases only) and $120 - 10x \ge 0$ gives $x \le 12$: **domain $0 \le x \le 12$**. $R(0) = 960$ and $R(12) = 0$, with a maximum of 1000: **range $0 \le R(x) \le 1000$**.	**[1]** (a) Price $8 + x$ **[1]** (a) Bowls $120 - 10x$ **[1]** (a) Product set up as R(x) **[1]** (a) Expanded: $-10x^2 + 40x + 960$ **[2]** (b) Vertex $x = 2$ **[1]** (b) Price $10 **[2]** (b) Maximum revenue $1000 (reports R, not x) **[1]** (b) Units **[2]** (c) Sets $R(x) \ge 960$ and simplifies **[2]** (c) $0 \le x \le 4$ **[2]** (c) Converts to prices $8 to $12 **[1]** (d) $x \ge 0$ and $x \le 12$ from bowls ≥ 0 **[1]** (d) Domain stated **[2]** (d) Range 0 to 1000 *[n] = points*
S4 APPLY	Consider the parabola $y = x^2 - 2x - 3$ and the line $y = x + 1$. **(a)** Solve the system algebraically. **(b)** Graph both and mark the solutions. **(c)** How many solutions does the system of the parabola and the line $y = -4$ have? And with $y = -6$? **(d)** Find k so that $y = 2x + k$ touches the parabola at exactly one point, and find that point.	**Answer:** (a) (−1, 0) and (4, 5) (c) $y = -4$: one solution, (1, −4); $y = -6$: no solutions (d) $k = -7$, point (2, −3) (a) $x^2 - 2x - 3 = x + 1$ gives $x^2 - 3x - 4 = 0$, so $(x-4)(x+1) = 0$. $x = 4$ gives $y = 5$; $x = -1$ gives $y = 0$: **(4, 5) and (−1, 0)**. (b) Parabola: vertex (1, −4), x-intercepts (−1, 0) and (3, 0). The line passes through (−1, 0) and (4, 5), so the graphs cross at the two solutions. (c) The vertex is the lowest point, $y = -4$. The line $y = -4$ touches it once: **one solution (1, −4)**. The line $y = -6$ is below the parabola: **no solutions**. (d) $x^2 - 2x - 3 = 2x + k$ gives $x^2 - 4x - 3 - k = 0$. One solution needs the discriminant $16 + 12 + 4k = 0$, so **$k = -7$**. Then $x^2 - 4x + 4 = 0$, $x = 2$, $y = -3$: **(2, −3)**. 	**[1]** (a) Sets the expressions equal **[1]** (a) Standard form $x^2 - 3x - 4 = 0$ **[2]** (a) Factors and finds $x = 4$ and $x = -1$ **[2]** (a) Finds both y-values (points, not just x) **[2]** (b) Parabola with its vertex **[1]** (b) Line $y = x + 1$ **[2]** (b) Both intersection points marked **[2]** (c) $y = -4$ has one solution (1, −4) **[2]** (c) $y = -6$ has no solution **[1]** (c) Reason based on the vertex **[1]** (d) Writes $x^2 - 4x - 3 - k = 0$ **[1]** (d) Discriminant equals 0 **[1]** (d) $k = -7$ **[1]** (d) Point (2, −3) *[n] = points*